Matrix-Free Second-Order Orbital Optimization for Valence Bond Self-Consistent Field Theory

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ABSTRACT

Orbital optimization in valence-bond self-consistent-field theory must accommodate nonorthogonal orbitals, prescribed atomic-orbital supports, and relaxation of the structure coefficients. We formulate a matrix-free second-order treatment in support-preserving nonredundant coordinates and distinguish the relaxed-orbital and coupled orbital–structure models. The numerical study will compare truncated Newton and norm-extended optimization with conventional and block-preconditioned L-BFGS, reporting convergence, operator work, wall time, and peak memory under matched conditions. [Replace this sentence with the principal quantitative findings and the supported conclusion after the benchmark campaign.]

1 INTRODUCTION

1.1 The optimization problem

Explain why VBSCF uses nonorthogonal localized orbitals and, in HAO calculations, fixed orbital supports. Separate the cost of optimizing orbitals from the cost of solving the structure-coefficient problem. Introduce OEO as full-AO support, not as an orthogonal-orbital model.

1.2 Relation to second-order MCSCF optimization

Review restricted-step Newton, augmented-Hessian and norm-extended optimization, and matrix-free CASSCF implementations [1–3]. Identify what carries over to VBSCF—the local coupled quadratic model and iterative operator actions—and what does not carry over without derivation: orthogonal orbital rotations and their redundancy rules.

1.3 Scope of this work

State the methodological contribution as a second-order optimization framework for the specified nonorthogonal VB variational space. Treat integral representation and string contraction as supporting implementations, not independent novelty claims in this paper. The principal numerical question is when additional curvature information improves convergence enough to repay its cost relative to block-LBFGS.

2 THEORY

2.1 Stationary structure problem and state averaging

Define the generalized structure eigenproblem at fixed orbital coefficients x. Use a different symbol for the AO overlap and the structure overlap throughout the manuscript.

$$\mathbf{H}_{\mathrm{VB}}(\mathbf{x})\mathbf{a}_s = E_s\mathbf{S}_{\mathrm{VB}}(\mathbf{x})\mathbf{a}_s, \quad \mathbf{a}_r^T\mathbf{S}_{\mathrm{VB}}\mathbf{a}_s = \delta_{rs} \tag{1}$$

$$\mathcal{E}(\mathbf{x}) = \sum_{s\in\mathcal{A}} w_s E_s(\mathbf{x}), \quad w_s = 1/m \tag{2}$$

For the single-state case, use the lowest stationary eigenvalue; for equal-weight state averaging, define the selected eigenspace and its separation from excluded roots. Derive normalization and selected-subspace gauge projections before introducing response equations.

2.2 Support-preserving nonredundant orbital coordinates

$$\mathbf{c}_p = \mathbf{R}_p\mathbf{x}_p, \quad \mathbf{x}(\mathbf{p}) = \mathcal{R}_k(\mathbf{p}), \quad \mathcal{R}_k(\mathbf{0}) = \mathbf{x}_k \tag{3}$$

Define the support injection $\mathbf{R}_p$, horizontal lift $\mathbf{U}_k$, local update map, and the physical step metric $\mathbf{M}_o$. Classify a direction as redundant only if it leaves the represented wave function unchanged within the selected structure space. In particular, active–active mixing is not automatically redundant for an arbitrary selected-VB expansion.

$$\mathbf{g}_o = \nabla_{\mathbf{p}}(\mathcal{E}\circ\mathcal{R}_k)(\mathbf{0}), \quad \mathbf{H}_o\mathbf{v} = D^2(\mathcal{E}\circ\mathcal{R}_k)(\mathbf{0})[\mathbf{v}] \tag{4}$$

Give the derivative of the actual update map, rather than transferring an orbital-rotation Hessian to a coefficient update. Check the consistency of lift, norm, and retraction.

2.3 Orbital–structure coupling and the relaxed Hessian

At a structure-stationary accepted point, write the constrained local Hessian in orbital and projected structure tangent variables. A is the fixed-structure orbital block, B is the mixed derivative, and C is the structure tangent block. All blocks include the chosen normalization and generalized-overlap constraints.

$$\mathbf{z} = \begin{pmatrix}\mathbf{p}\\ \mathbf{q}\end{pmatrix}, \quad \mathbf{g} = \begin{pmatrix}\mathbf{g}_o\\ \mathbf{0}\end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix}\mathbf{A} & \mathbf{B}^T\\ \mathbf{B} & \mathbf{C}\end{pmatrix} \tag{5}$$

$$\mathbf{H}_{\mathrm{rel}} = \mathbf{A} - \mathbf{B}^T\mathbf{C}^\dagger\mathbf{B}, \quad \mathbf{q}_N = -\mathbf{C}^\dagger\mathbf{B}\mathbf{p}_N \tag{6}$$

Define $C^\dagger$ only on the admissible response space. Explain the equivalence of the unshifted coupled Newton and relaxed-orbital Newton limits, and why shifted coupled and eliminated models need not be identical away from that limit.

2.4 Restricted steps and residual certification

$$m(\mathbf{z}) = \mathcal{E}_k + \mathbf{g}^T\mathbf{z} + \tfrac{1}{2}\mathbf{z}^T\mathbf{K}\mathbf{z}, \quad \mathbf{z}^T\mathbf{M}\mathbf{z} \le \Delta_k^2 \tag{7}$$

$$\mathbf{r} = (\mathbf{K} + \lambda\mathbf{M})\mathbf{z} + \mathbf{g}, \quad \lambda \ge 0, \quad \mathbf{K} + \lambda\mathbf{M} \succeq 0 \tag{8}$$

$$\lambda(\mathbf{z}^T\mathbf{M}\mathbf{z} - \Delta_k^2) = 0, \quad \|\mathbf{r}\|_{\mathbf{M}^{-1}} \le \epsilon_{\mathrm{abs}} + \eta_k\|\mathbf{g}\|_{\mathbf{M}^{-1}} \tag{9}$$

Separate stationarity, step feasibility, complementarity, and curvature checks. State the regularity and inner-solve conditions required for superlinear or quadratic local convergence; an HVP implementation alone does not establish those properties.

2.5 TNHVP and norm-extended optimization

Describe TNHVP as a truncated Newton realization of the relaxed model, and NEO as a norm-extended formulation of the local second-order model [1]. Define the gradient scaling and the step extracted from the generalized augmented eigenvector explicitly.

$$\begin{pmatrix} 0 & \alpha\mathbf{g}^T \\ \alpha\mathbf{g} & \mathbf{K} \end{pmatrix}\begin{pmatrix} \beta \\ \mathbf{y} \end{pmatrix} = \mu\begin{pmatrix} 1 & 0 \\ 0 & \mathbf{M} \end{pmatrix}\begin{pmatrix} \beta \\ \mathbf{y} \end{pmatrix} \tag{10}$$

$$\mathbf{z} = \mathbf{y}/(\alpha\beta), \quad \lambda = -\mu, \quad (\mathbf{K} + \lambda\mathbf{M})\mathbf{z} = -\mathbf{g} \tag{11}$$

Discuss the nonzero homogeneous component, lowest-root selection, and the interior Newton limit. Describe how the projected solver reaches this model; do not equate a generic augmented-Hessian step with a certified norm-constrained solution.

2.6 Preconditioning and controlled structure response

Distinguish the Hessian model from the preconditioner. Present the one-electron block preconditioner, retained secant information where applicable, and the role of coupled structure response. If the implementation omits or restricts response according to a work estimate, identify that calculation as an approximate model and report it separately.

3 IMPLEMENTATION

3.1 Matrix-free derivative actions

Describe the accepted-point data required for gradient and HVP evaluation: normalization, inactive projection, integral transformation, structure action, and response. Identify reused quantities and directional quantities, without claiming that matrix-free means no large intermediates. Give memory costs in orbital, structure, and unique-string dimensions.

3.2 Trial steps and accepted-point updates

$$\rho_k = \frac{\mathcal{E}_k - \mathcal{E}(\mathcal{R}_k(\mathbf{p}))}{m(\mathbf{0}) - m(\mathbf{z})} \tag{12}$$

Explain trial evaluation, acceptance, trust-radius updates, and rebuilding the local model. State which quantities are refreshed at a keyframe and which are reused during subspace expansion. Count rejected trials as work even when they do not advance the accepted-iteration counter.

Algorithm 1. Accepted-point second-order VBSCF optimization (outline)

1. Read the same orbital guess and structure list used by all comparison methods.
2. Solve the selected structure problem; evaluate energy and the physical orbital gradient.
3. Build the support-preserving local coordinates, metric, derivative actions, and preconditioner.
4. Expand the iterative second-order model; certify its residual and record response treatment.
5. Retract the orbital step, solve the trial structure problem, and evaluate model agreement.
6. Accept or reject the trial; refresh accepted-point data and test both stopping criteria.

3.3 Cost accounting

Report orbital HVPs, orbital-to-structure and transpose-coupling actions, structure H/S actions, trial evaluations, and accepted steps separately. Include setup, final observables, and total process memory in end-to-end measurements; isolate optimizer time in an additional breakdown.

4 COMPUTATIONAL DETAILS

4.1 Methods and matched conditions

Method	Selector	Role
XMVB-style L-BFGS	ISCF=5	Conventional quasi-Newton baseline
TNHVP	ISCF=7	Matrix-free truncated Newton comparison
Block-LBFGS	ISCF=8	Enhanced baseline with one-electron block initial inverse

Method	Selector	Role
NEO	ISCF=9	Norm-extended second-order method

Table 1. Comparison methods. Input selectors are implementation details, not method names.

Use identical initial orbitals, geometry, basis, active electrons/orbitals, spin, selected structure list, state weights, integral representation, and final convergence tests. Compare exact and RI only in separate matched groups. Dense and Davidson structure solvers must solve the same selected generalized eigenproblem to controlled accuracy.

Record the projected-gradient convention and the physical dual-metric norm. The working energy-change and projected-gradient targets are 10^{-7} hartree and 10^{-3}, respectively; confirm the precise implemented norms and final settings before collection. Use an independently tighter calculation for endpoint comparisons.

4.2 Test-set design

Group	Examples	Purpose
Small derivative oracles	F_2; small selected-structure cases	Finite differences, dense Hessian, endpoint checks
Localized HAO	241 (benzene), 240, 7975, 7963, YAMSAI	Support constraints and convergence variation
Transition metals	MnF_2 and $FeCl_2$	Conditioning, spin, and OEO/HAO behavior
Larger response spaces	CERRAS and LOFLEA	Response cost and memory
Equal-weight state average	Small multistate cases; selected pyrrole/OEO cases	Root-space and response stability

Table 2. Proposed test groups; molecular specifications and eligibility remain to be frozen.

Do not include unresolved endpoint discrepancies as successful validation. A PySCF CASSCF comparison is a variational-space reference only when the OEO structure expansion, spin/symmetry sector, integrals, and core convention match; otherwise it is a diagnostic comparison.

4.3 Hardware and reproducibility

Collect publication timings on Hanhai25 compute nodes through Slurm. Report CPU model, allocated cores, affinity, OpenMP/BLAS threads, compiler flags, dependency versions, and commit identifier. Record peak RSS at the job/process level. Preserve nonconverged runs and resource failures in the dataset rather than dropping them.

5 RESULTS AND DISCUSSION

5.1 Derivative and response validation

Table 3 will report gradient directional differences, HVP directional differences, bilinear symmetry defects, and coupled-versus-Schur agreement on small cases. Include nonorthogonal well-conditioned and near-singular orbital points, HAO and OEO, and at least one equal-weight state average. [Insert verified numerical errors.]

5.2 Local convergence

Figure 2 will show energy error and physical gradient norm for identical controlled perturbations around a common stationary point. Establish the local convergence regime separately from the number of iterations required to enter it. [Insert trajectories and the residual forcing sequence.]

5.3 End-to-end comparison

Figures 3 and 4 will plot accepted steps and total wall time against a common system ordering, with failures visibly marked. Table 4 will provide final energies, energy changes, physical gradients, action counts, wall time, and RSS. Discuss cases where fewer steps do not yield lower wall time. [Insert matched results; do not mix revisions.]

5.4 Where the second-order cost is spent

Figure 5 will break down setup, core HVP, coupling/structure response, trial evaluation, and final output. Analyze larger active spaces using measured work and memory rather than accepted-step count alone. Compare full and restricted response with the model treatment stated for every run. [Insert stage timings and response statistics.]

5.5 Ablations and limitations

Separate model changes from preconditioner changes. Prioritize matched full/restricted response and one-electron block/Hessian-diagonal comparisons; identity preconditioning need not be a production benchmark. Discuss small structure gaps, incomplete selected-VB spaces, root changes, and redundant-coordinate conditioning. [Insert only ablations that answer a specific algorithmic question.]

6 CONCLUSIONS

Summarize the validated second-order formulation, the measured range of efficiency, and the cost of structure relaxation. State which method is recommended for which workload only if the data support that distinction. [Write the quantitative conclusion after Sections 5.1–5.5 are complete.]

ASSOCIATED CONTENT

Supporting Information

Planned contents: coordinate and response derivations; update-map derivatives; finite-difference protocols; complete molecular specifications and initial orbitals; raw convergence trajectories; non-convergence records; timing scripts and environment details. Submit the SI separately from the main manuscript.

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DATA AVAILABILITY STATEMENT

[Supply a persistent public repository link or DOI for the input files, initial orbitals, raw outputs, benchmark metadata, and analysis scripts. Identify the code release and commit used for every figure. Explain any justified restrictions.]

REFERENCES — STARTER LIST

(1) Jensen, H.-J. Aa.; Jørgensen, P. A Direct Approach to Second-Order MCSCF Calculations Using a Norm Extended Optimization Scheme. *J. Chem. Phys.* **1984**, *80* (3), 1204–1214.

(2) Werner, H.-J.; Knowles, P. J. A Second Order Multiconfiguration SCF Procedure with Optimum Convergence. *J. Chem. Phys.* **1985**, *82* (11), 5053–5063.

(3) Sun, Q.; Yang, J.; Chan, G. K.-L. A General Second Order Complete Active Space Self-Consistent-Field Solver for Large-Scale Systems. *Chem. Phys. Lett.* **2017**, *683*, 291–299.

[Complete the VBSCF/XMVB, inexact-Newton, trust-region, L-BFGS, and density-fitting literature before submission. This starter list is not an exhaustive priority review.]